

ELEC-H415

Modulation and Coding Project Report

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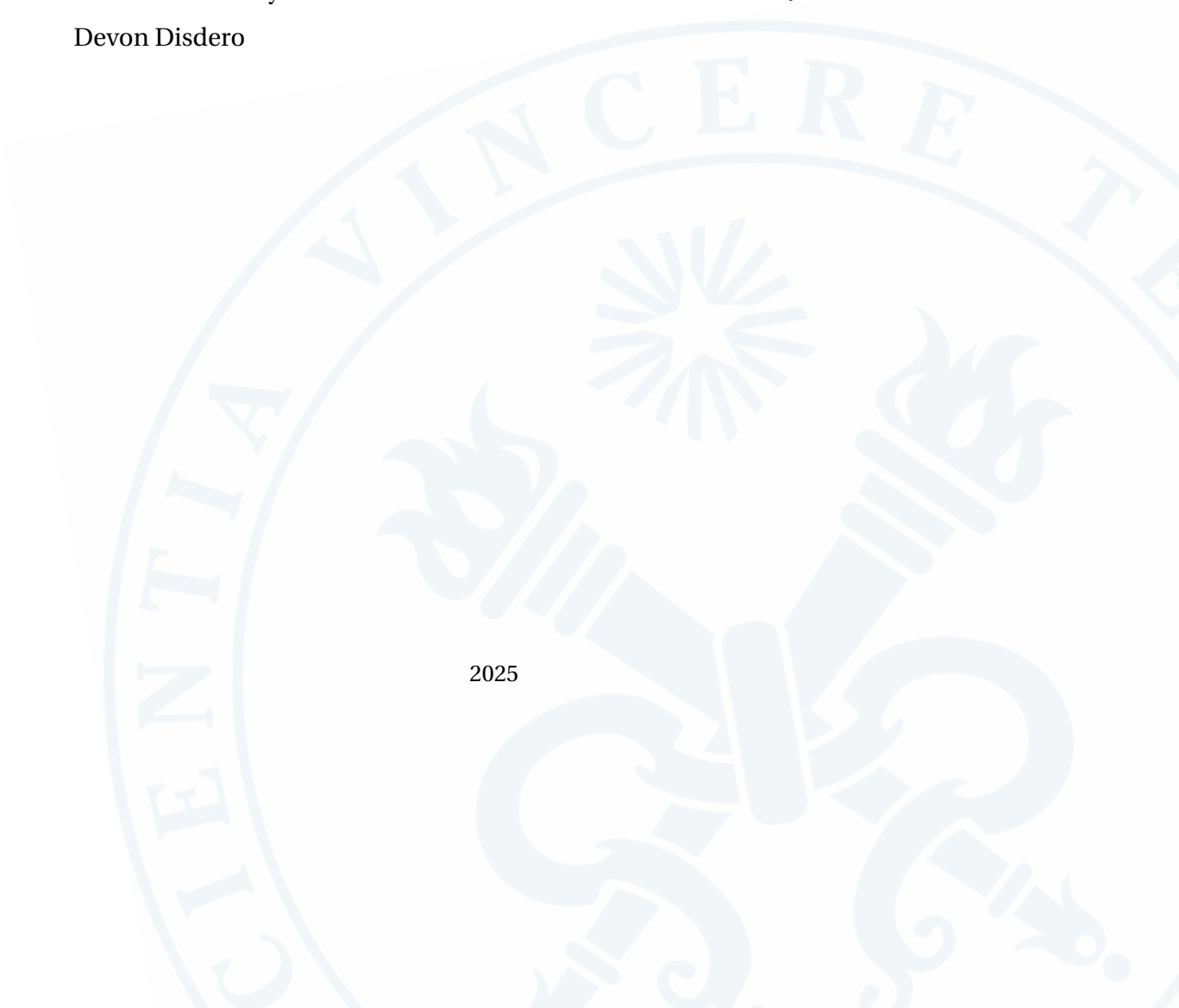
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Contents

1	Introduction	1
2	System Overview and Objectives	1
3	Communication Chain Implementation	1
3.1	Symbol Mapping and Demapping	1
3.2	Pulse Shaping and Matched Filtering	2
3.3	Noise Modeling and BER Estimation	2
4	Time and Frequency Synchronization	3
4.1	Timing Offset Correction - Gardner Algorithm	3
4.2	Frame and Carrier Frequency Acquisition	4
4.3	Simulation Results for Synchronization	7
5	MATLAB Code Structure and Execution	8
5.1	Primary Files	8
5.2	Execution Flow	9
6	Simulation Results	9
6.1	BER vs. E_b/N_0	9
7	Conclusion	10
8	Questions from Orange Seminar and Visite	10
8.1	HFC architecture and its main components	10
8.2	HFC network evaluation in near future	11
8.3	Orange network typical incidents	12
8.4	Orange Data Center Characteristics	12

1 Introduction

The goal of this project is the design and simulation of a DVB-C (Digital Video Broadcasting for Cable) communication chain in MATLAB. Encompassing various elements of digital communication, it includes symbol mapping, pulse shaping, synchronization, and error correction coding. The main objective is to model a realistic transmission chain over a Hybrid Fiber-Coax (HFC) infrastructure, in order to analyze resulting behavior under different channel impairments.

2 System Overview and Objectives

The communication chain is representative of a real DVB-C system. It includes:

- A modulation and demodulation subsystem
- A baseband channel simulation with AWGN
- Time and frequency synchronization mechanisms
- Channel encoding/decoding using Low-Density Parity Check (LDPC) codes

The project is structured in three major phases:

1. Optimal communication chain implementation
2. Synchronization algorithms (Gardner timing recovery, frame/CFO acquisition)
3. LDPC coding and decoding

Each phase is evaluated with simulation results in MATLAB.

3 Communication Chain Implementation

3.1 Symbol Mapping and Demapping

The process begins by mapping binary input bits to complex modulation symbols using high-order QAM. The function `mapping.m` is used to perform this task based on constellation order (e.g., 16-QAM or 64-QAM). At the receiver, `demapping.m` performs the inverse operation using minimum distance criteria.

3.2 Pulse Shaping and Matched Filtering

The transmitter applies a half-root Nyquist filter using `root_cos_fil_gen.m`. This limits spectral occupancy and minimizes ISI. The receiver employs a matched filter with the same characteristics. Together, they form a Nyquist filter pair that theoretically eliminates ISI when sampled correctly.

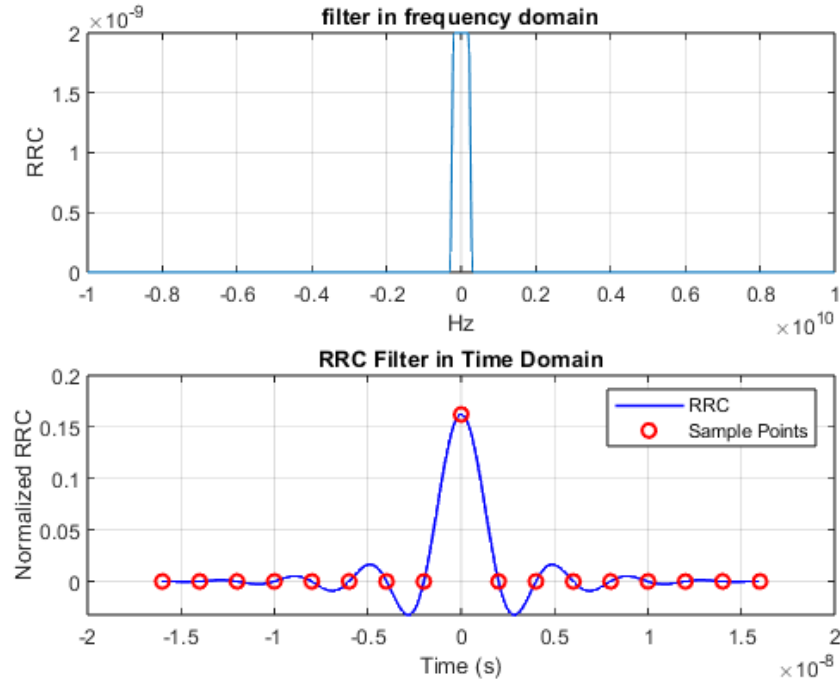


Figure 1: RRC filter time and frequency domain

3.3 Noise Modeling and BER Estimation

The AWGN model is introduced by adding white Gaussian noise with a controlled power level according to desired E_b/N_0 . Simulation of BER performance over multiple E_b/N_0 values is performed and compared to theoretical curves.

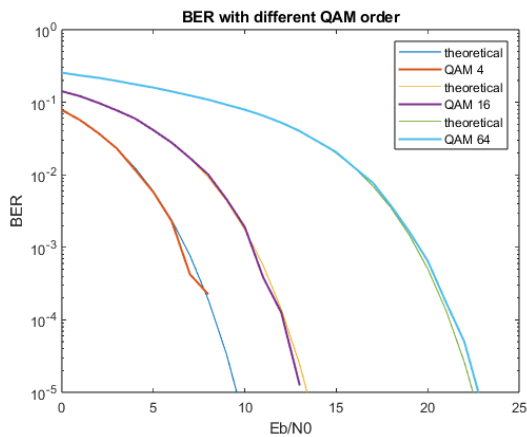


Figure 2: BER of noisy signal

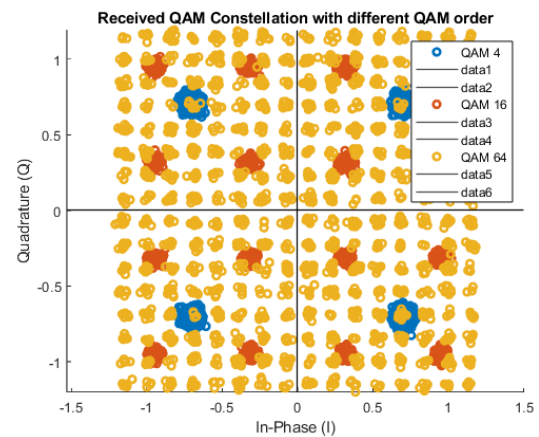


Figure 3: IQ constellation of noisy signal

4 Time and Frequency Synchronization

4.1 Timing Offset Correction - Gardner Algorithm

Timing synchronization is addressed first using the Gardner algorithm, implemented in `gardner_function.m`. It computes timing error estimates and performs feedback correction. Simulation results show good convergence and robustness to CFO.

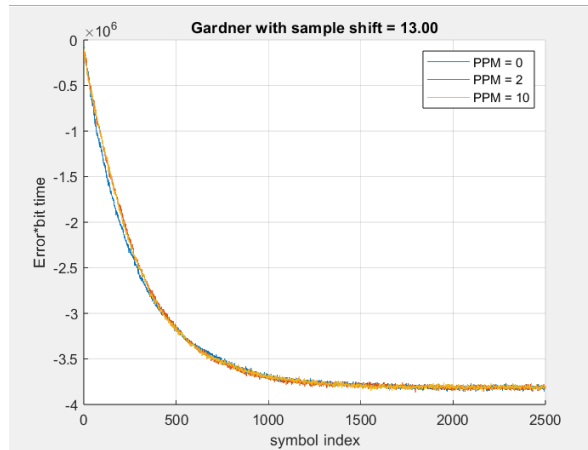


Figure 4: Gardner algorithm with different CFO (13 sample shifts)

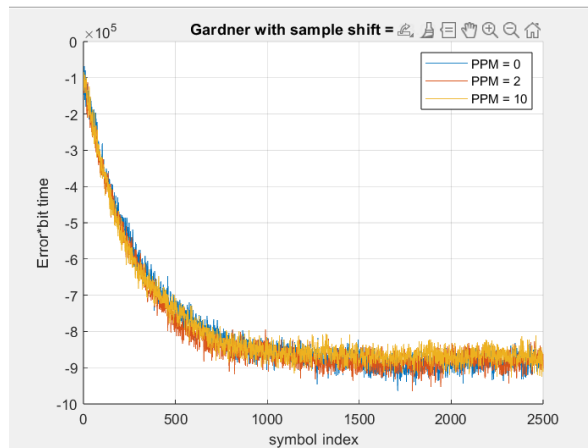


Figure 5: Gardner algorithm with different CFO (23 sample shifts)

The Gardner algorithm relies on detecting the optimal sampling instants by estimating timing errors using the interpolated midpoint sample and its adjacent symbols. It operates in a feedback loop, updating the sampling offset based on the observed error at each step.

The gain parameter K plays a critical role in the convergence behavior of the Gardner loop:

- If K is too small, the algorithm converges slowly. It may require many symbols to correct timing misalignment, which increases latency.

- If K is too large, the performance of the algorithm is degraded given that it becomes more sensitive to noise, which may exhibit oscillations or divergence.
- An optimal K ensures a balance between fast convergence and stable tracking, even in the presence of channel noise.

To visualize the influence of K , we plot the timing error trajectory for different values of K . This illustrates how quickly and smoothly the algorithm adapts to the correct sampling position under varying feedback strengths.

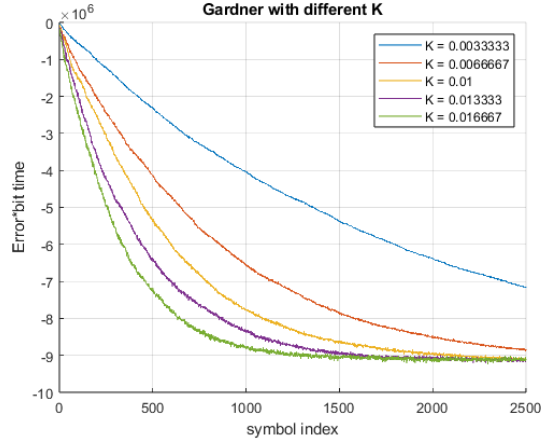


Figure 6: Gardner algorithm for different K

4.2 Frame and Carrier Frequency Acquisition

Frame and carrier frequency acquisition is crucial for aligning the receiver's frame with the incoming signal and correcting any frequency offsets. In our implementation, this is handled using a differential cross-correlation approach in the function `dataAcquisition.m`. This technique helps to detect the start of frame (TOA) and estimate the carrier frequency offset (CFO) even in the presence of noise.

1. Impact of pilot length on acquisition accuracy: We evaluate the effect of varying pilot lengths on the estimation accuracy. Longer pilots generally provide more reliable correlation, reducing the variance of both time and frequency error estimates.

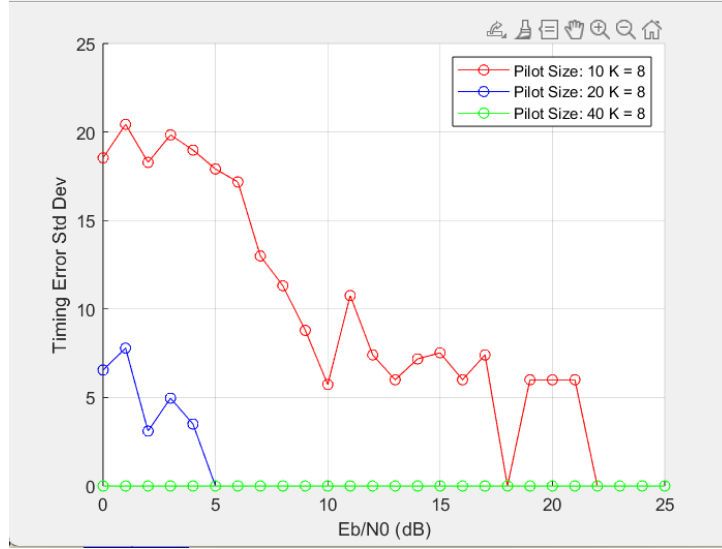


Figure 7: Time error variances relative to different pilot lengths. Window size $K = 8$

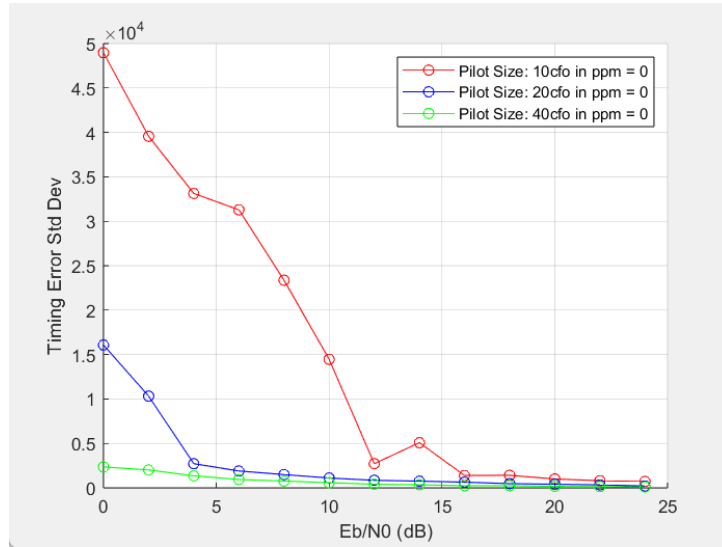


Figure 8: Frequency error variances relative to different pilot lengths. Window size $K = 8$

2. Impact of the parameter K on acquisition: Here, we assess how changing the parameter K , affects acquisition feedback sensitivity, thereby influencing performance. A properly tuned K yields faster convergence without instability, while extremes may produce oscillation or slower response.

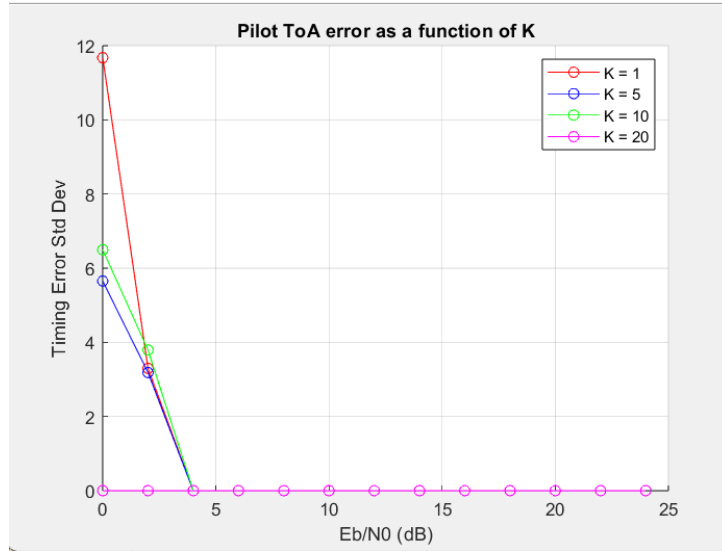


Figure 9: Time error variances relative to parameter K . **Pilot length = 20**

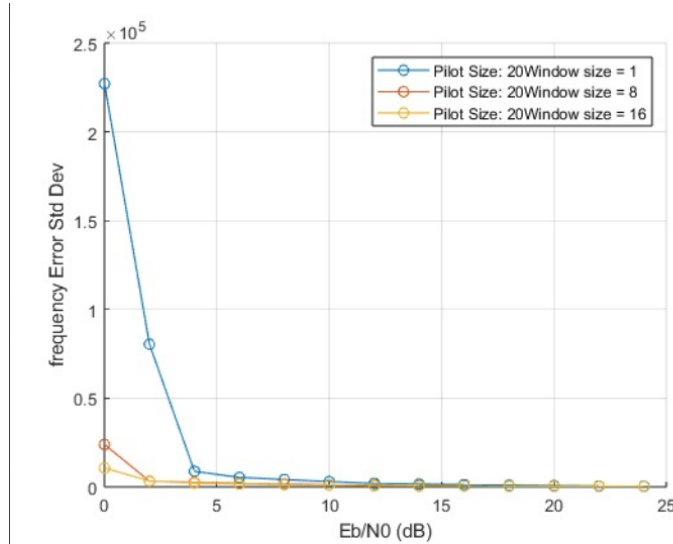


Figure 10: Frequency error variances relative to parameter K . **Pilot length = 20**

3. Acquisition under modified CFO: Finally, we take an optimized pilot length and K value, then apply different CFO values to test acquisition robustness. This step illustrates how resilient the acquisition system is when operating under varying real-world impairments.

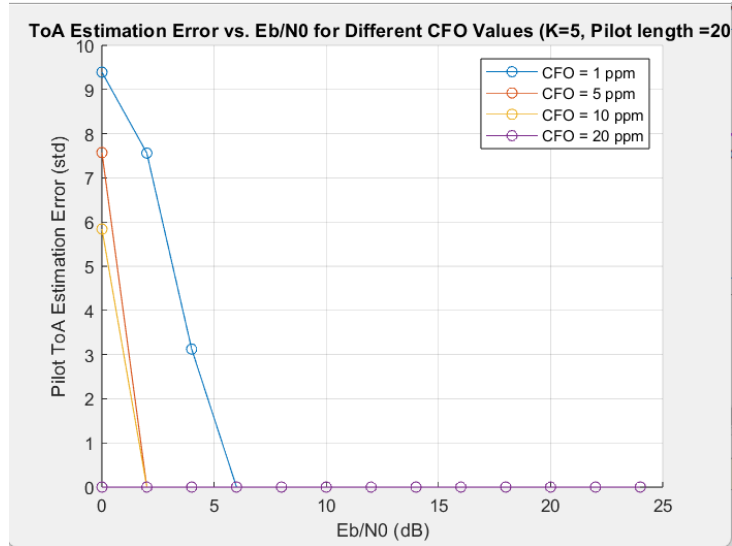


Figure 11: Toa error standard deviation for modified CFO with fixed pilot length = 20 and K= 5

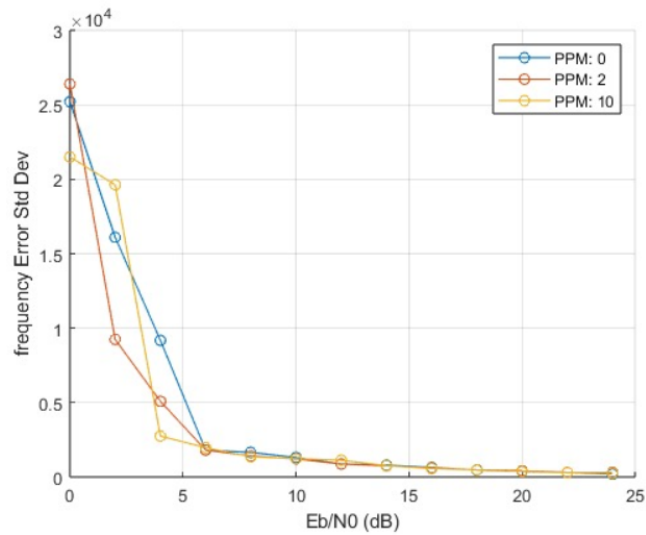


Figure 12: Frequency err for modified CFO, with fixed **pilot length** = 20 and **K=10**

4.3 Simulation Results for Synchronization

- **Impact of CFO:** BER vs. CFO simulations show linear degradation up to a threshold.

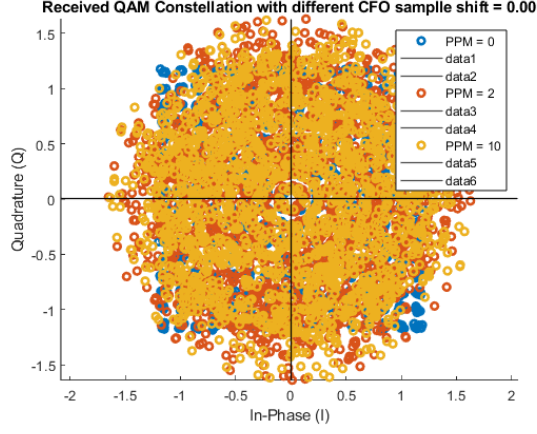


Figure 13: CFO effect on BER

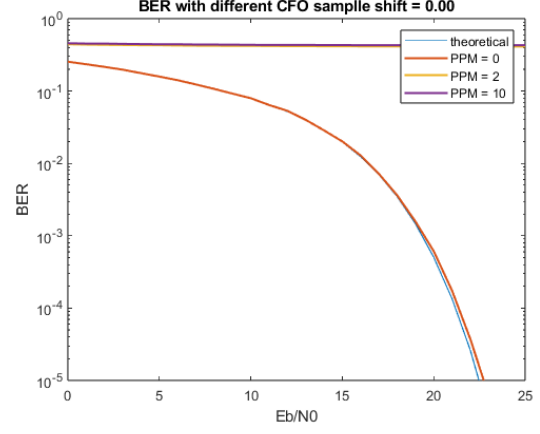


Figure 14: CFO effect on IQ

- **CFO correction by hand:** BER vs. CFO by hand correction simulations show remaining ISI effect.

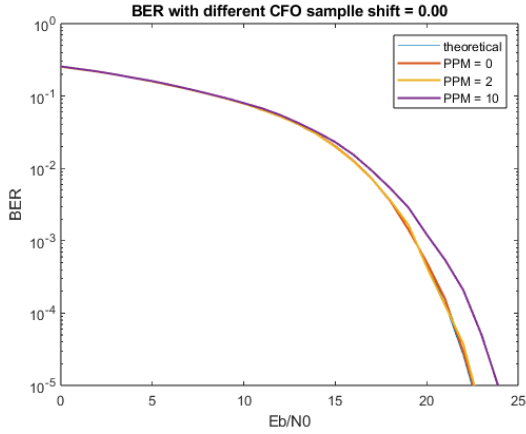


Figure 15: canceled CFO effect on BER

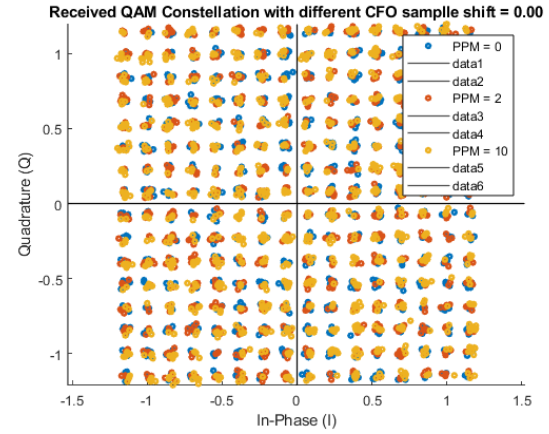


Figure 16: canceled CFO effect on IQ

- **Frame Acquisition Accuracy:** Performance is tested for different pilot lengths and values of K.

5 MATLAB Code Structure and Execution

5.1 Primary Files

- `dataAcquisition.m` — data acquisition logic
- `mapping.m`, `demapping.m` — mapping bitstream into modulation techniques
- `gardner_function.m`, `interpolate.m` — Timing synchronization
- `main.m`, — main script which use above mention scripts to generate mapp filter and demap

5.2 Execution Flow

The simulation begins by generating random bits, followed by symbol mapping, filtering, noise addition, synchronization correction, and BER evaluation. Multiple test scripts investigate performance under different impairments.

6 Simulation Results

6.1 BER vs. Eb/N0

The simulated BER closely matches the theoretical curves for 16-QAM and 64-QAM.

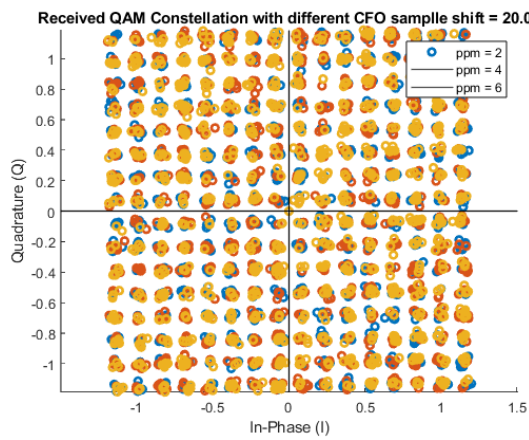


Figure 17: BER of simulated result

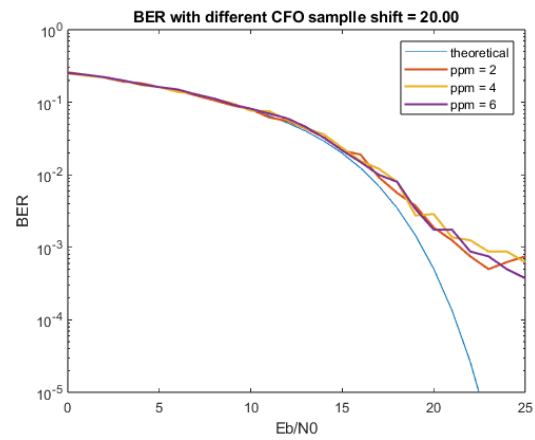


Figure 18: IQ of simulated result

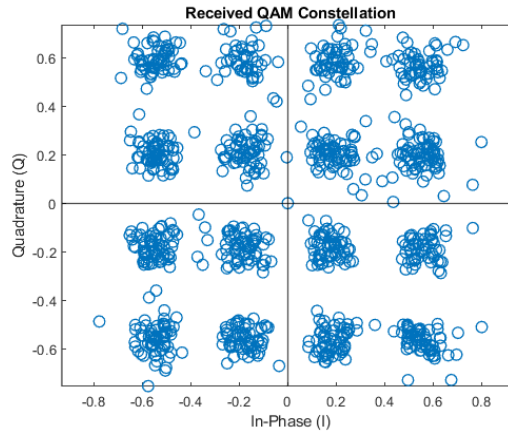


Figure 19: IQ of PLUTO result

Also see the recording of the IQ graph result using the wired PLUTO device via the following link:
IQ graph recording using wired PLUTO device.

And the recording of the IQ graph result using the wireles PLUTO device via the following link:
IQ graph recording using wireles PLUTO device.

7 Conclusion

This report presented a full DVB-C baseband simulation in MATLAB. It included modulation, filtering, synchronization, and LDPC error correction. The implementation was validated by simulation of BER, timing recovery, and decoder convergence. The modularity of the MATLAB scripts also enables future expansion, such as using the Adalm-Pluto SDR for over-the-air tests.

References

- ELEC-H401 Course Notes: Francois Horlin
- MATLAB Communications Toolbox
- DVB-C and DOCSIS Standards

8 Questions from Orange Seminar and Visite

8.1 HFC architecture and its main components

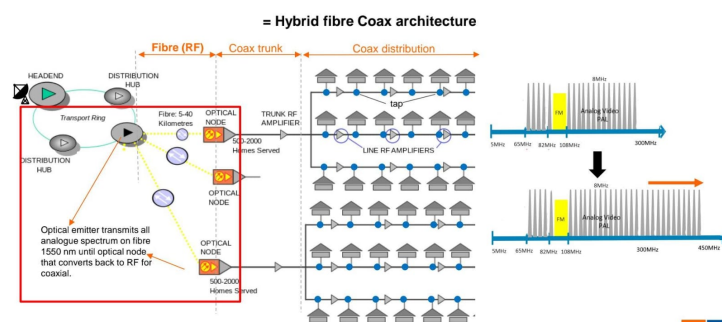


Figure 20: HFC architecture and main components

The current capacity bottleneck in HFC networks lies primarily in the upstream (US) direction. As shown in the theoretical bandwidth allocation:

- **Upstream (US):** Limited to 150 MHz, supporting only 1–2 Gbps using OFDMA.
- **Downstream (DS):** Allocated 1000 MHz, allowing up to approximately 10 Gbps using OFDM.

This mismatch in available bandwidth poses a constraint on upstream capacity, seeing as modern applications now require higher upload speeds. As such, despite a rather large downstream capacity, the limited upstream bandwidth is the main bottleneck in current HFC deployments.

8.2 HFC network evaluation in near future

While Hybrid Fibre-Coaxial (HFC) is not being abandoned, it is evolving in parallel with Fibre To The Premises (FTTP). Upcoming advancements aim to:

- Support higher bandwidths
- Lower latency
- Improve energy efficiency and reduce maintenance

Key Technologies Driving the Evolution

DOCSIS 4.0

- **Extended Spectrum DOCSIS:** Increases spectrum from 1.2 GHz to 1.8 GHz.
- **Full Duplex DOCSIS (FDX):** Allows both upstream and downstream communication in the same frequency band at the same time.
- Requires new amplifiers, better shielding, and mechanisms to cancel power.

Distributed Access Architecture (DAA)

- **Remote PHY/Remote MAC-PHY:** Moves digital processing closer to users.
- Replaces RF transmission over fibre with Ethernet/IP, improving SNR, which reduces power, and increases modularity.

Node Splitting & Fibre Deepening

- Reduces the number of homes per optical node
- Increases bandwidth per user
- Enables better QoS during peak usage

FTTP Integration

- Operators like Orange are heavily investing in XGS-PON and 50G-PON.
- Supports transition or supplementation of HFC with full-fibre solutions.
- Enables shared passive infrastructure with open-access for service providers.

Advanced Modems & Wi-Fi 6/7

- Support multi-gigabit speeds
- Mesh capabilities for in-home distribution

8.3 Orange network typical incidents

It is mainly due to construction works, such as road maintenance or urban development activities, during which digging operations accidentally damaging underground cables and infrastructure. These physical disruptions can cause temporary service interruptions until repairs are completed which is taking at least a few hours.

For critical locations such as hospitals, emergency services, and data centers, redundant lines or backup connections are often installed to ensure continuous connectivity. For high priority objects such as mentioned above the reconstruction works are being done in faster temp as no one wants to have a problem in reaching emergency contacts.

8.4 Orange Data Center Characteristics

The Orange data center is designed for a maximum power capacity of 3 MW, but currently operates at approximately 1.5 MW. It utilizes a combination of 400 V AC and 48 V DC power distribution systems. The data center employs passive cooling techniques, with six cooling blocks located on the rooftop. But has additional space reserved for future expansion if higher cooling capacity is required. For backup power, the facility is equipped with two generators. Each is approximately rated at 750 kW. In addition, there is a fuel tanker that is located in the parking area. It can supply energy for up to 72 hours, in case of emergency, thereby ensuring that it functions during power outages.

4.5 Questions and Answers

Regarding the Simulation

Q1: It is proposed to use the baseband equivalent model of the AWGN channel. Would it be possible to work with a bandpass implementation of the system?

A: Yes, it is possible to work with a bandpass implementation, but it is not necessary for most simulations. Baseband models are preferred due to their simplicity and computational efficiency. They focus on the complex envelope of the signal and eliminate the need to simulate the carrier frequency, making analysis and processing easier.

Q2: How do you choose the sample rate in MATLAB?

A: The sample rate should satisfy the Nyquist criterion and is typically defined as:

$$f_s = R_s \cdot L$$

where R_s is the symbol rate and L is the oversampling factor. This rate ensures the accuracy of the time-domain representation and adequate filtering.

Specifically in the case of QAM modulation, R_s is defined by:

$$R_s = \frac{R_b}{\log_2 M} \quad \text{where } M \text{ is modulation order}$$

The sample rate becomes:

$$f_s = \frac{R_b}{\log_2 M} \cdot L$$

Q3: How do you make sure you simulate the desired E_b/N_0 ratio?

A: The noise power is calculated from:

$$N_0 = \frac{E_b}{10^{E_b/N_0 \text{ (dB)}/10}}$$

In MATLAB, the AWGN is added as:

```
noise_power = Eb / (10^(EbNO_dB/10));  
noise = sqrt(noise_power/2) * (randn(size(s)) + 1i*randn(size(s)));  
received = s + noise;
```

E_b is derived from the average symbol energy divided by the number of bits per symbol.

Q4: How do you choose the number of transmitted data packets and their length?

A: It depends on:

- Desired accuracy (e.g., simulate until 100 errors are seen).
- Target BER range (lower BERs require more bits).
- Computation time and memory constraints.

A typical simulation transmits millions of bits for accurate BER estimation.

Regarding the Communication System

Q5: Determine the supported (uncoded) bit rate as a function of the physical bandwidth.

A: The bit rate R_b is given by:

$$R_b = R_s \cdot \log_2(M)$$

Assuming Nyquist signaling:

$$R_s = 2B \Rightarrow R_b = 2B \cdot \log_2(M)$$

where B is the bandwidth and M is the constellation size.

Q6: Explain the trade-off communication capacity/reliability achieved by varying the constellation size.

A: Increasing M (constellation size) improves spectral efficiency (more bits per symbol) but reduces reliability due to closer symbol spacing, making it more sensitive to noise. Thus:

- Higher M = Higher capacity, lower reliability
- Lower M = Lower capacity, higher reliability

Q7: Why do we choose the half-root Nyquist filter to shape the complex symbols?

A: The half-root Nyquist (e.g., root raised cosine) filter meets the Nyquist criterion when applied at both transmitter and receiver. The product of two half-root filters results in a full Nyquist filter, eliminating ISI while offering smoother transition bands and easier implementation. In addition, the matched filter results in the property that SNR is maximized.

Q8: How do we implement the optimal demodulator? Give the optimisation criterion.

A: The optimal demodulator in AWGN is a matched filter followed by maximum likelihood detection:

$$\hat{s} = \underset{s \in \mathcal{S}}{\operatorname{argmin}} \|r - s\|^2$$

where r is the received signal and $\mathcal{S}$ is the set of constellation points.

One interpretation is that we are trying to minimize the euclidean distance between the received signal and constellation points.

Q9: How do we implement the optimal detector? Give the optimisation criterion.

A: The optimal detector uses the maximum likelihood (ML) criterion:

$$\hat{b} = \underset{b}{\operatorname{argmax}} P(r|b)$$

In AWGN with equally likely bits, this simplifies to minimizing Euclidean distance between the received signal and transmitted symbol. For coded systems, MAP or sequence detection (e.g., Viterbi) may be used.

5.6 Questions and Answers

Regarding the Simulation

Q1: Derive analytically the baseband model of the channel including the synchronisation errors.

A: The received baseband signal $r(t)$, when affected by carrier frequency offset (CFO), carrier phase offset, and timing offset is:

$$r(t) = s(t - \tau) \cdot e^{j(2\pi\Delta f t + \phi)} + n(t)$$

where:

- τ : timing offset (sampling time shift)
- Δf : carrier frequency offset
- ϕ : carrier phase offset
- $n(t)$: additive white Gaussian noise

This model is commonly used in baseband simulations to include synchronization impairments.

Q2: How do you separate the impact of the carrier phase drift and ISI due to the CFO in your simulation?

A: CFO introduces both a *phase drift* and a *rotating constellation* over time. The *ISI effect* emerges due to misalignment in symbol timing over multiple symbols. To separate them:

- Simulate CFO without filtering to observe only phase drift (no ISI).
- Simulate CFO with pulse shaping and matched filtering to capture ISI effects.

Comparing these two helps isolate ISI from phase drift.

Q3: How do you simulate the sampling time shift in practice?

A: Sampling time shift is simulated by offsetting the sampling instant in MATLAB:

```
rx_shifted = interp1(t, signal, t + delta_t, 'linear', 0);
```

or, in the discrete domain, by circular shifting or fractional delay filters to mimic the effect of timing mismatch.

Q4: How do you select the simulated E_b/N_0 ratio?

A: We choose a desired range of E_b/N_0 in dB (e.g. 0-20 dB), and then compute the corresponding power of the noise:

$$N_0 = \frac{E_b}{10^{E_b/N_0 \text{ (dB)}/10}}$$

We adjust the variance of AWGN added to the signal by using the energy per bit.

Q5: How do you select the lengths of the pilot and data sequences?

A: Pilot length is chosen to balance estimation accuracy and overhead. Long pilots improve synchronization but reduce throughput. Data length is chosen based on:

- Statistical reliability of BER
- Simulation time constraints
- The number of bits needed to observe target performance metrics

Regarding the Communication System

Q6: In which order are the synchronisation effects estimated and compensated. Why?

A: The typical order is:

- (a) Sampling time instants are acquired and tracked, using Gardner algorithm
- (b) Frame starting time is acquired
- (c) Carrier frequency offset estimation
- (d) Phase drift compensation (due to remaining CFO) via interpolation between pilot sequences

This order avoids ambiguity since frame and frequency misalignments can distort timing error calculations if not corrected early. The remaining phase drift, on the other hand, can be compensated for at the end.

Q7: Explain intuitively how the error is computed in the Gardner algorithm. Why is the Gardner algorithm robust to CFO?

A: Gardner uses interpolated mid-sample values to estimate timing error:

$$e(n) = \Re \{ [r(nT + T/2) - r(nT - T/2)] \cdot r^*(nT) \}$$

It compares the slope between two samples and the reference to determine whether to speed up or slow down the sampling clock. It is robust to CFO because it uses the envelope of the signal, not the phase, making it less sensitive to carrier frequency rotation.

Q8: Explain intuitively why the differential cross-correlator is better suited than the usual cross-correlator? Isn't it interesting to start the summation at $k = 0$ (no time shift)?

A: Differential cross-correlation uses the change between symbols, reducing the sensitivity to **unknown phase offsets**. It is more robust against phase noise and CFO. Starting at $k = 0$ includes correlation without delay, which may contain autocorrelation noise. Skipping $k = 0$ focuses on actual differential structure and **removes DC bias**.

Q9: Are the frame and frequency acquisition algorithms optimal? If yes, give the optimisation criterion.

A: In theory, it is optimal if designed based on **maximum likelihood (ML)** or **maximum a posteriori (MAP)** principles.

- Frame acquisition is optimal if it maximizes the probability of correct symbol boundary detection.
- Frequency acquisition is optimal if it minimizes the frequency estimation error (e.g., ML estimate of Δf from pilot correlation).

In practice, implementations are sub-optimal due to complexity and limited pilot information.